1. **A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.**

Step	What to Do	What You Are Checking For
1. Check the Power Adapter and Cable	Ensure the charger is securely connected to both the laptop and the wall outlet. If possible, test with another compatible charger.	To determine whether the charger, cable, or power outlet is faulty and not supplying power to the laptop.
2. Inspect the Battery	If the laptop has a removable battery, remove it and try powering on using only the AC adapter. If the battery is internal, check for charging indicator lights.	To identify whether a faulty or completely discharged battery is preventing the laptop from turning on.
3. Perform a Hard Reset	Disconnect the charger, remove the battery (if removable), hold the power button for 15–30 seconds, then reconnect the charger and try again.	To discharge residual electrical charge that may be preventing the laptop from starting.
4. Check for Power Indicators	Press the power button and observe LED lights, charging lights, fan movement, or listen for beep codes.	To determine whether the laptop is receiving power and identify possible motherboard or hardware issues based on indicator behavior.
5. Check the RAM	Turn off the laptop, reseat the RAM module(s), or test with one RAM stick at a time if multiple modules are installed.	To verify that loose or faulty RAM is not preventing the system from completing the Power-On Self-Test (POST).

2. **You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.**

Troubleshooting Flow: Blank Monitor (PC Appears On)

Follow these structured steps from easiest to hardest to isolate the failure.

1. Check Monitor Power and Status

- **Verify power:** Ensure the monitor power LED light is on.
- **Toggle power button:** Turn the monitor off and back on.
- **Check the outlet:** Plug the monitor directly into a known working wall outlet.
- **Observe the LED color:** Note if the light is solid, blinking, amber, or blue.

2. Verify Video Cable Connections

- **Check connection seat:** Unplug and firmly re-seat both ends of the video cable.
- **Verify port placement:** Ensure the cable connects to the dedicated **GPU ports** (lower down), not the motherboard ports (higher up), unless using integrated graphics.
- **Test alternative cables:** Swap out the current HDMI or DisplayPort cable for a known working one.
- **Isolate input source:** Press the "Input" button on the monitor to manually select the correct active port (e.g., HDMI 1).

3. Isolate the Display Hardware

- **Test another display:** Connect the PC to a different monitor or a TV.
- **Test another device:** Connect a laptop or console to the original monitor.
- **Disconnect peripherals:** Unplug all USB devices except the keyboard and mouse to rule out power-draw issues.

4. Diagnose Internal PC Components

- **Perform a hard reset:** Power down, unplug the PC wall cord, hold the PC power button for 30 seconds, plug back in, and boot.
- **Check Motherboard Debug LEDs:** Look inside the case for lit **POST LEDs** (marked CPU, RAM, VGA, BOOT) indicating a hardware failure.
- **Reseat the GPU:** Power off, remove the dedicated graphics card, clean the slot, and firmly click it back into place.
- **Reseat the RAM:** Remove all memory modules and test booting with only **one RAM stick** placed in the primary slot.

3. **A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.**

Isolating an Unresponsive Wireless Mouse

Follow this systematic process to isolate the failure between the hardware, port, or software settings.

1. Test the Mouse (Hardware Isolation)

- **Check the power:** Flip the mouse over to verify the power switch is in the **ON** position.
- **Observe the sensor:** Check if the optical LED light underneath the mouse is illuminated.
- **Replace the battery:** Insert a fresh, brand-new battery to eliminate low-voltage connectivity drops.
- **Test on another device:** Connect the mouse receiver to a different laptop or PC. If it works there, the mouse hardware is perfectly fine.
- **Verify connection type:** For Bluetooth mice, press the dedicated **sync/pairing button** on the bottom to re-initiate communication.

2. Test the USB Port (Interface Isolation)

- **Switch ports:** Move the USB wireless dongle to a different port on the PC.
- **Avoid USB hubs:** Plug the dongle directly into the motherboard or case ports, bypassing any unpowered hubs or monitor pass-throughs.
- **Cross-test with the keyboard:** Unplug the functional keyboard and plug the mouse dongle into that exact, verified working USB port.
- **Clear interference:** Move any high-bandwidth USB 3.0 devices (like external hard drives) away from the wireless dongle to prevent signal crowding.

3. Test System Settings (Software Isolation)

- **Navigate with keyboard:** Use the Windows Key, Tab, and Arrow Keys to open **Device Manager**
- **Check device status:** Expand the **Mice and other pointing devices** section to check for a yellow warning triangle or a red 'X'.
- **Reinstall drivers:** Highlight the mouse driver, press Shift + F10 (or the Menu key) to open the context menu, select **Uninstall device**, and restart the PC to let Windows reinstall it automatically.
- **Disable power saving:** Open the properties of the **USB Root Hub** inside Device Manager, go to **Power Management**, and uncheck **"Allow the computer to turn off this device to save power."**

4. **Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.**

Top Hardware Causes for Gaming PC Overheating

- In a high-utilization environment like a gaming cafe, hardware components are pushed to their limits. Here are the three most likely culprits for the thermal shutdowns and how to inspect them:

1. Clogged Cooling Fins and Fans (Dust Accumulation)

- Gaming cafe PCs pull in massive volumes of air, rapidly trapping dust, lint, and hair inside the heatsinks.
- **How to check:** Power off the PC, open the side panel, and inspect the CPU cooler, GPU fans, and case intake filters. Look for dense mats of dust blocking the metal cooling fins. Manually spin the cooling fans with your finger while the system is off to ensure they rotate freely and smoothly without grinding or resisting.

2. Dried or Degraded Thermal Paste

- Thermal paste transfers heat from the processor to the cooler. Over time—especially under constant, high-temperature gaming cycles—it dries out, cracks, and acts as a thermal insulator instead of a conductor.
- **How to check:** Boot the PC and open a hardware monitoring tool (like HWMonitor or Open Hardware Monitor). If the CPU temperature instantly spikes to 90°C–100°C the moment a game launches, the thermal transfer is failing. To confirm physically, remove the CPU cooler; dried paste will appear chalky, flaky, or completely evaporated.

3. Failed Liquid Cooling Pump (AIO Failure)

- If the gaming rig utilizes an All-In-One (AIO) liquid cooler, the mechanical pump inside the block can burn out, or the liquid inside can permeate and evaporate over years of heavy use.
- **How to check:** Power on the PC and gently place two fingers on the rubber tubes leading from the CPU block to the radiator. You should feel a slight, steady vibration or fluid movement. Alternatively, enter the system BIOS and check the **CPU_FAN** or **AIO_PUMP** RPM readings. If the readout shows 0 RPM while the PC is melting down, the pump motor is dead.